

OPSIMIUS

Promoting green hydrogen for a sustainable future

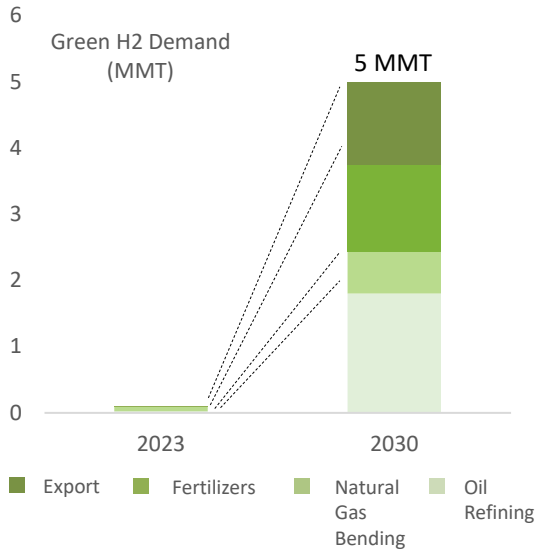
TEAM STREET BOYZ
PRANSHI GARG | KRITIKA GUPTA | LAKSHAY NARANG

BRIEF OVERVIEW

Outline of the green hydrogen industry, challenges faced and the present hydrogen value chain

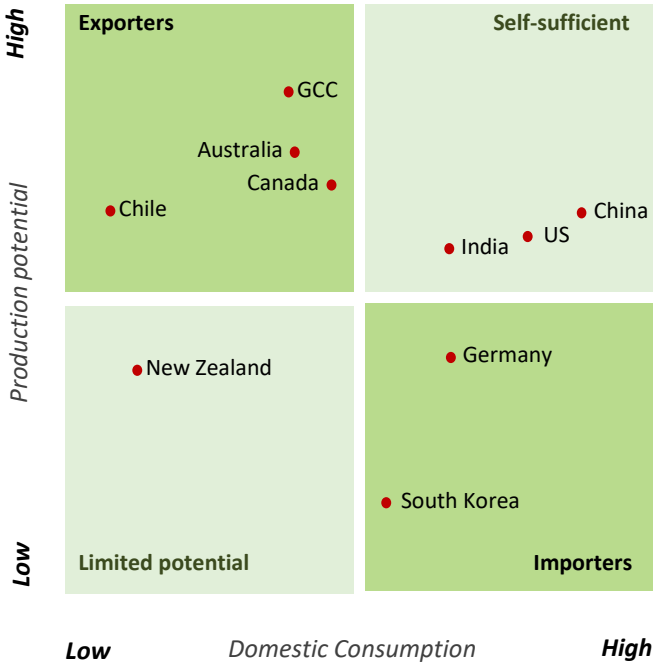
INDUSTRY ANALYSIS

Green hydrogen demand till 2030 as per pathway to NHM target assumptions



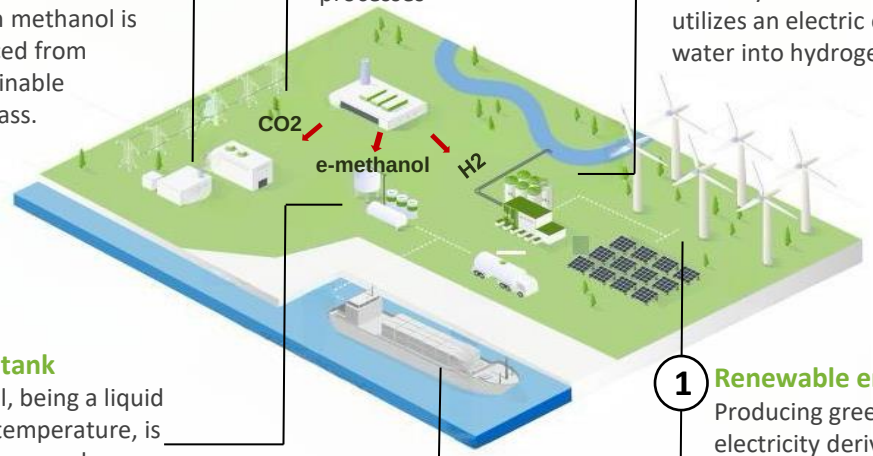
96 MMT ***CO2 emissions avoided by 2030***
Refer to appendix

with GCC countries showing great export potential..



CURRENT SUPPLY CHAIN

The green hydrogen value chain, from production to consumption, is composed of multiple elements..



4 Biomass power plant
The carbon dioxide required to produce green methanol is sourced from sustainable biomass.

3 Methanol synthesis plant
Methanol is synthesised & distilled by chemical processes

2 Electrolyser
Electrolysis is a chemical process that utilizes an electric current to split water into hydrogen and oxygen

5 Storage tank
Methanol, being a liquid at room temperature, is easy to store and transport.

1 Renewable energy
Producing green hydrogen requires electricity derived from renewable energy sources.

6 Uses of green methanol
Green hydrogen serves as a chemical feedstock and fuel, with great potential in shipping.

Challenges faced by the Green Hydrogen Industry…..

Lack of domestic demand hinders investment

India needs investments in green hydrogen along with **long-term offtake contracts of more than 15+ years.**

Cost Barriers

High costs such as **USD 4-5** per kg hydrogen impacts it's economic viability affecting **production, storage and transportation costs**

Inadequate power system structures

Electrolysers face **short-term load generation imbalances** but can earn **57%** more by providing ancillary services to power grids.

Uncertain future demand and technical standards

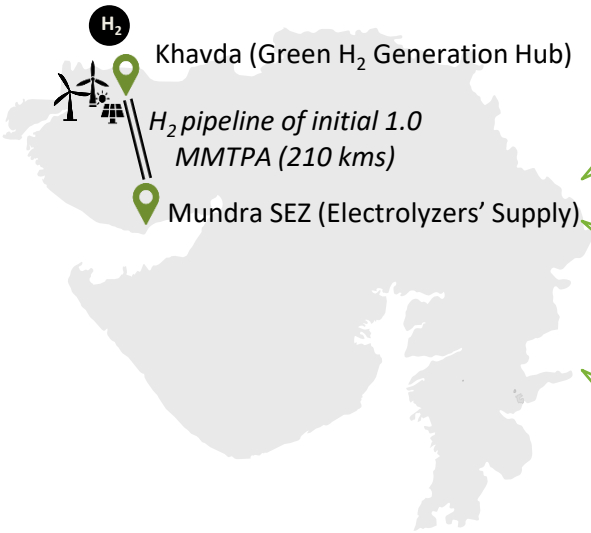
Green Hydrogen's infancy stage and absent technical standards hinder green hydrogen's growth.

Source: McKinsey

SOURCING AND PRODUCTION

Powering Decarbonization: Strategies for Sourcing Renewables and Producing Green Hydrogen

GREEN HYDROGEN GENERATION



- Khavda site (~**71,000 acres**) in Gujarat with large scale renewable deployment potential of ~**20 GW**

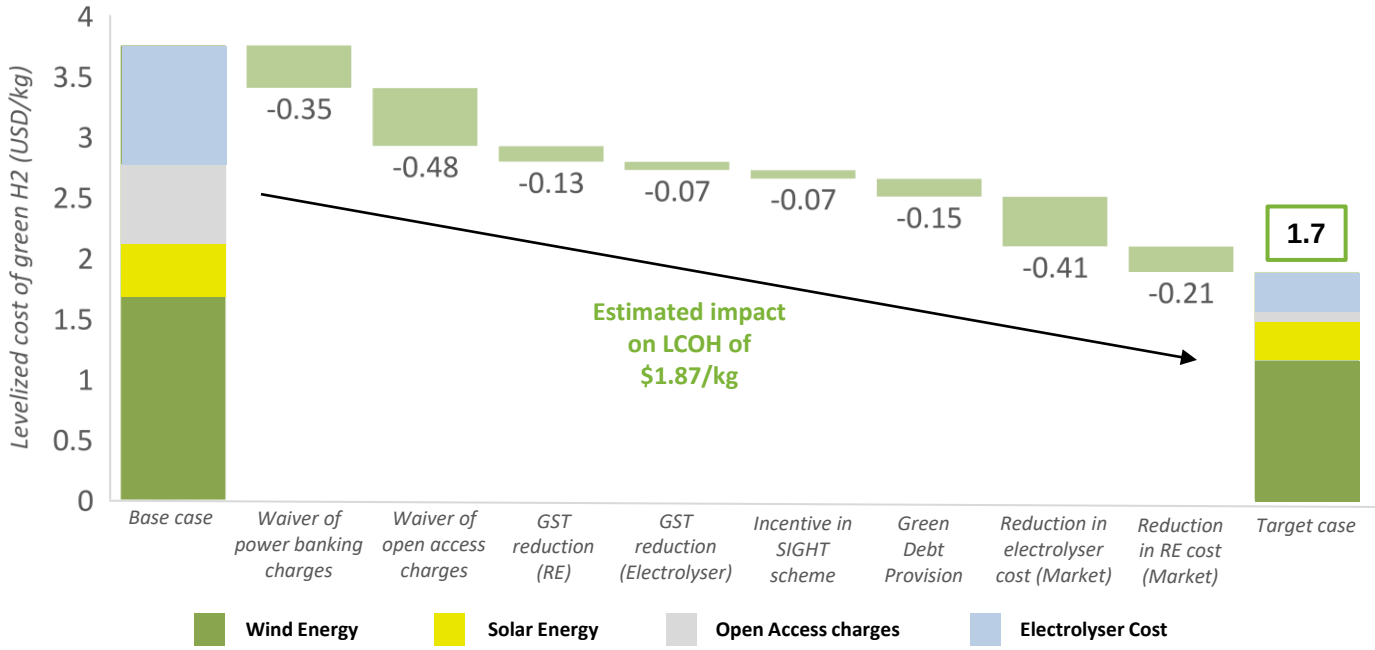
- Electrolyser to be sourced from manufacturing facility at Mundra SEZ

- Green H_2 generation plant to be located near the renewable cluster. Green H_2 to be transported via pipeline to Mundra

THE TRI-PARTITE AGREEMENT

Objective	*Partners*	*Rationale*
Wind & Solar Energy Supply	ReNew POWER	✓ **US$5.3 billion** - MoU with **ADB** ✓ **~8.3 GW** Capacity
Ports and Electrolyzers' Hub	MUNDRA PORT SPECIAL ECONOMIC ZONE	✓ Modern **Port** facility ✓ **SEZ** - Government incentivized
Hydrogen Production Plants	adani	✓ **USD 50 bn** investment by **2030** ✓ **Khavda -** Largest power plant

REDUCTION IN THE OVERALL PRODUCTION COST



1 Setting up Power Banking

Renewable energy banking could help in reducing the cost of green H2 by **increasing the utilisation of electrolysers**

2 Low-cost open-access power

Waivers on **interstate open-access charges** could support the provision of wheeling power between **special economic zones**

3 Reducing GST

Electrolysers are taxed at **18%**, and renewable energy setups at **13.8%**. The reduction in GST could be planned.

4 Reducing Capital Costs

Providing direct fiscal support through low-**cost loans, government-owned institutions**, and other forms of financial assistance.

Impact

3B Impact on Govt finances by 2030

Electrolysers

Alkaline Based | **PEM Based** | **Solid Oxide**

→ *Refer to Appendix*

Source: Economic Times

STORAGE AND TRANSPORTATION

Potential hydrogen carriers and rationale behind them

REASONS BEHIND SELECTING E-METHANOL

Globally traded methanol ensures safe, simple handling with minimal incidents

BIOMETHANOL

COST EFFECTIVE

Methanol can deliver hydrogen at **50% cost** of producing hydrogen through electrolysis.

LOW CARBON DENSITY

Compared to conventional fuels, methanol reduces nitrogen oxide emissions by up to **60%,** and decarbonizes industries

ENERGY EFFICIENT

Electrolysis consumes 50-55 kWhr electricity/kg h2 whereas methanol reforming consumes **<0.04 kWhr electricity/kg h2** that yields high energy.

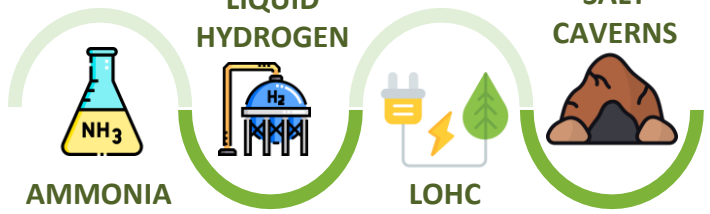
RENEWABLE POWER INTERMITTENCY

Power-to-liquid stores excess renewable energy as methanol, boosting renewable asset value and lowering investment risk.

EASE OF STORAGE AND HANDLING

Methanol is liquid at room temperature along with a low freezing point of **– 97°C**, making methanol fuel cells suitable for use in cold ambient temperatures.

Other Hydrogen Carriers



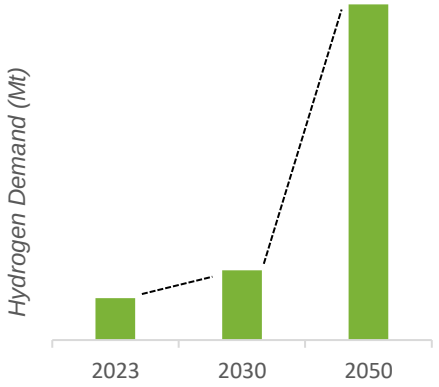
Despite gaseous hydrogen's cost-effectiveness, liquid hydrogen remains the most viable carrier due to lack of transportation infrastructure for gaseous hydrogen.

Comparison- delivery of 45MJ of energy

Parameters	E-methanol	Lead acid Batteries	Hydrogen
Weight	5.5 KG	230 KG	63 KG
Volume	7L	96L	50L

Production Units which covers major portion of Methanol Demand

due to which demand for Methanol is expected to grow by 65% by 2030



PATHWAY TOWARDS IMPLEMENTATION

Methanol Technologies

DMFC
Produces electric power by the direct conversion of the methanol fuel

RMFC
Produces hydrogen to be reacted in the fuel cell to produce electricity.

Uses of E-methanol

- **Transportation** using methanol fuel cells
- Used as **feedstock** like biomass or coal
- Production **of green fuels**
- Indirectly helps in **Grid Stabilization**

Potential Collaboration with Mitsubishi Gas Chemical Trading Company

Objective

- Supporting **regulatory compliance**
- Fostering **technology partnership**
- Establishing **methanol demand**

MITSUBISHI GAS CHEMICAL

Rationale

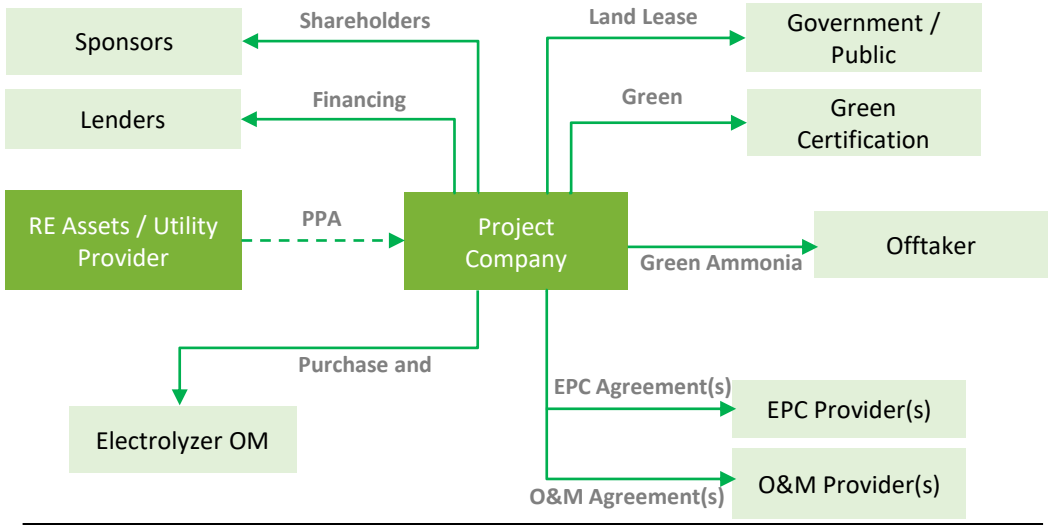
- ✓ Development of DMFC
- ✓ High Energy Density **(4.4 Kwh/L)**
- ✓ High Capacity (**1.6 Mn-4.05 Mn tonnes** per year)
- ✓ Portability

Source: methanol.org, Advent

FINANCING AND GOVERNMENT INTERVENTIONS

Empowering Green Hydrogen: Financial and Policy Solutions for India

Representative Green Hydrogen Project Structure



GREEN BOND

~7.29% *yield on 10-year SGBs*

~ ₹20K Cr *Worth GB to be issued in FY25*

- ✓ *Investor Confidence*
- ✓ *Positive Reputation*
- ✓ *Regulatory Alignment*

Export Credit Agency

ECAs back exports with loans, guarantees & insurance.

Potential ECA Collaborations

ECGC (A Government of India Enterprise)

Korea Eximbank — THE EXPORT-IMPORT BANK OF KOREA

Key Takeaways

- Lower Interest Rates
- Lower Repayment Terms
- Government Guarantees

Japan-India Joint Crediting Mechanism (JCM) for carbon credits

Carbon Credit Revenue: The credits can be sold to Japan, creating a new revenue stream for Indian companies.

Boosting Investment: The JCM will attract foreign investors, providing funding and accelerating technology development.

Export opportunities: JCM will give Indian companies a competitive edge in the global market and will attract buyers worldwide.

Technological Collaboration: Indian producers can access advanced hydrogen technology through partnerships with Japanese leaders.

UNDERSTANDING NGHM

- NGHM defines "green hydrogen" as having emissions below **2 kg** of **CO_2** equivalent per kg of H_2 produced over **12-months**.
- **Strategic Interventions for Green Hydrogen Transition (SIGHT) Programme** provides financial and promotional support.

GOVERNMENT BODIES

- **Bureau of Indian Standards (BIS)** issues standards for green hydrogen value chain in collaboration with **MNRE**.
- **Oil Industry Safety Directorate (OISD)** sets safety guidelines for storage and handling of hydrogen.

INVOLVED ORGANIZATIONS

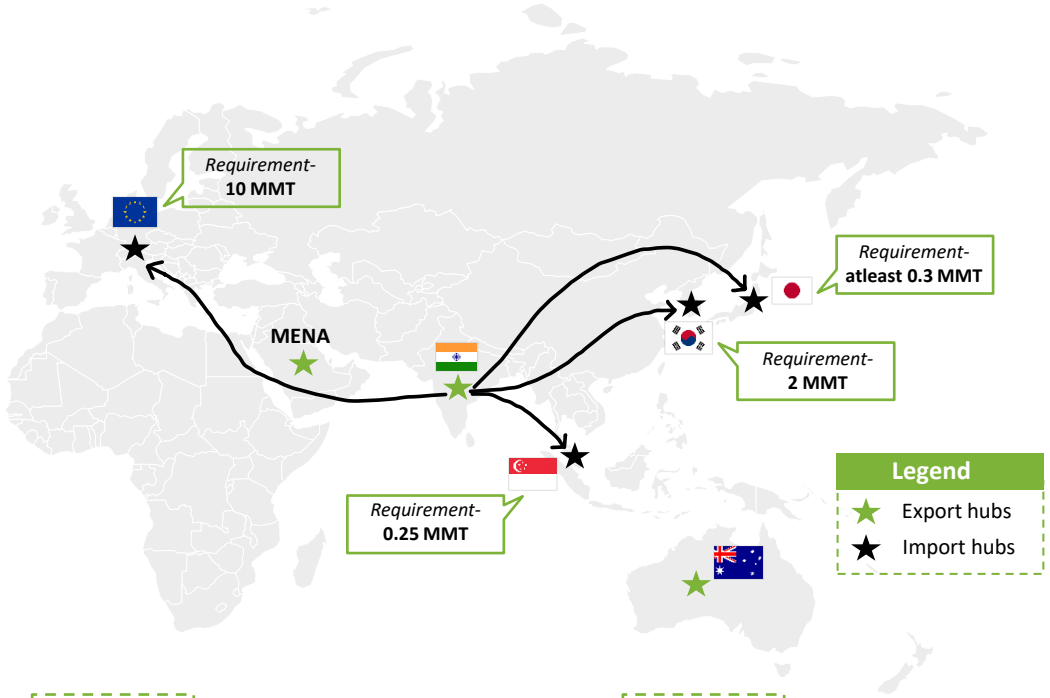
- **Petroleum and Explosives Safety Organisation** regulates the manufacture, storage, and transport of industrial explosives.
- **Petroleum and Natural Gas Regulatory Board (PNGRB)** governs the transportation and marketing of natural gas.

Source: World Economic Forum, Bain

EXPORTS

Empowering Global Decarbonization: Green Hydrogen Exports from India

POTENTIAL EXPORT DESTINATIONS



10% Target for achieving **World's Green hydrogen** market share by **2030** under NHM

12.5 Million tons — Green H2 demand by **EU, Japan, Korea & Singapore** by 2030

Ship Route

EUROPE	KOREA	JAPAN	SINGAPORE
9700 km	*11000 km*	*9760 km*	*3502 km*
Port de Marseille, France	Port Busan, Seoul	The Port of Osaka	Singapore Port

POTENTIAL EXPORT HUBS	Landed Cost of Hydrogen	POTENTIAL IMPORT HUBS			
		South Korea	EU	Japan	Singapore
INDIA	🟡	🟢	🟡	🟢	🟢
MENA	🟢	🟡	🟢	🔴	🟡
UNITED STATES	🟡	🟡	🟡	🟡	🔴
AUSTRALIA	🟢	🟢	🔴	🟢	🟡
LATAM	🟢	🔴	🔴	🔴	🔴

Key Takeaways

1. Target to meet **15%** of Japan's and Korea's import demands, **5%** of EU's and **50%** of Singapore's import demands by 2030
2. Other countries such as **Bangladesh and Sri Lanka** could become potential export destinations in the future

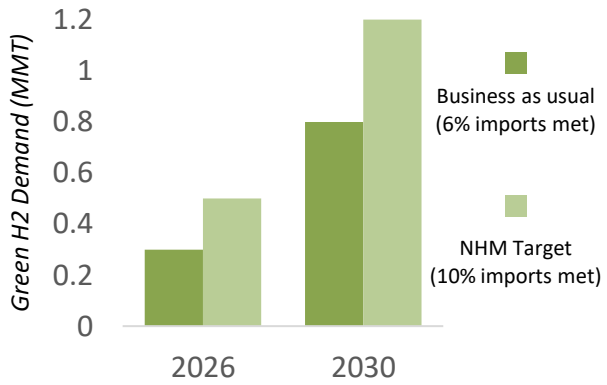
Securing Offtakers

IHI X ACME

The Indian Renewable energy company signed a pact with **Japanese industry** major to supply **green ammonia** to Japan

5B Dollars investment for 1.2 MMT plant

Green H2 Export Demand by 2030



IMPACT GENERATED

20 MMT carbon emissions avoided

$22 Billion investment need

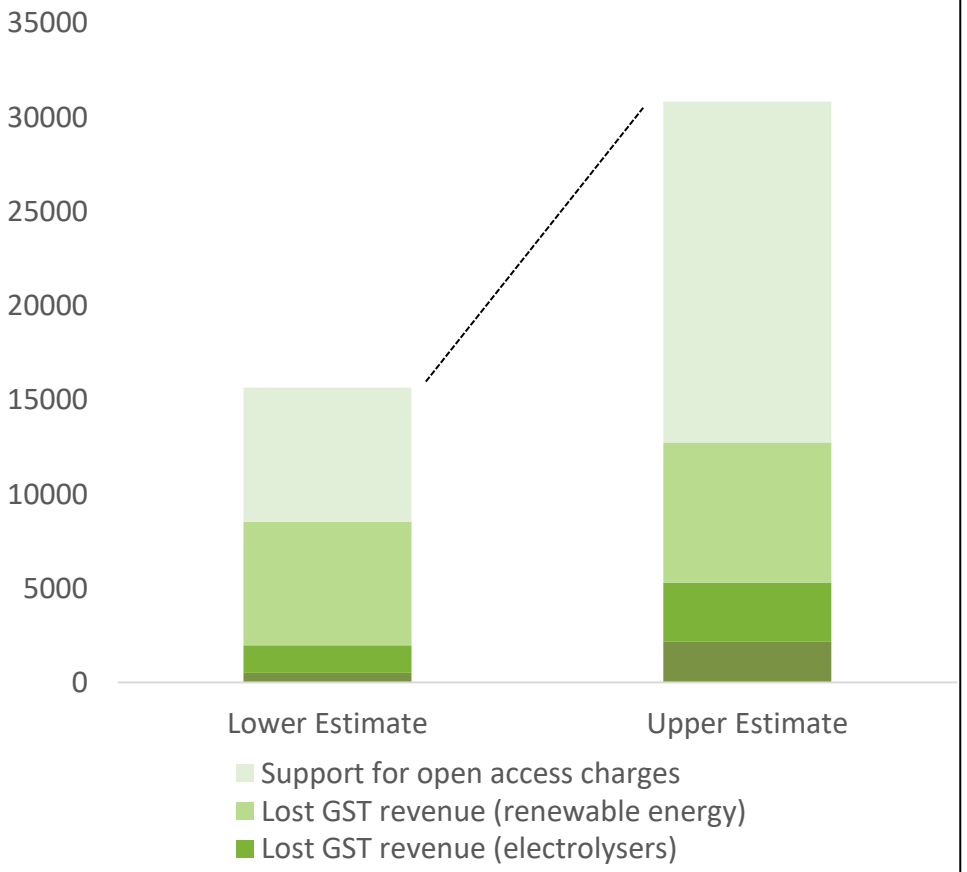
Source: USAID


APPENDIX

APPENDIX

EXHIBIT 1

Estimated Impact of the considerations on finances of state and central governments (INR crore)



Note: To simplify the calculation, we will use the average of the upper and lower estimates.

EXHIBIT 2

Technology and cost review of different electrolyser stacks

PARAMETERS	Alkaline Electrolyser	PEM Electrolyser	Solid Oxide Electrolyser
TRL	🟢	🟡	🔴
Conversion Efficiency	🟡	🔴	🟢
Life Span	🟢	🟢	🔴
Response Time (Fast & Slow)	🟡	🟢	🔴
Safety	🟡	🟢	🔴
CAPEX	🟢	🟡	🔴

EXHIBIT 3

Energy available along the conversion and transport chain in hydrogen equivalent terms, 2030…

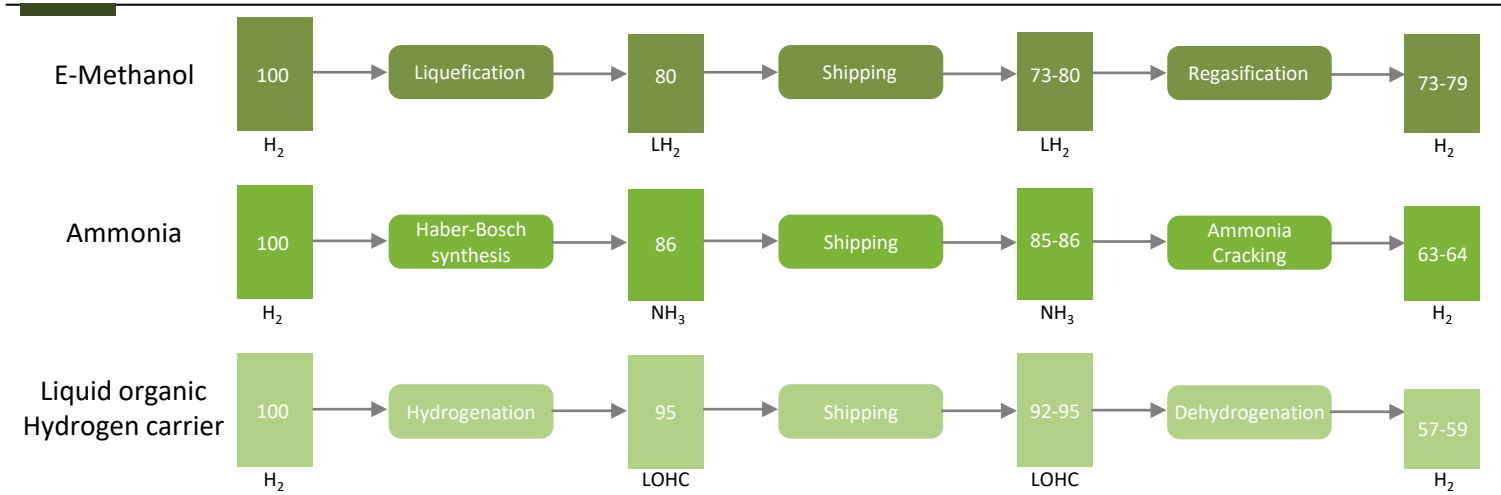
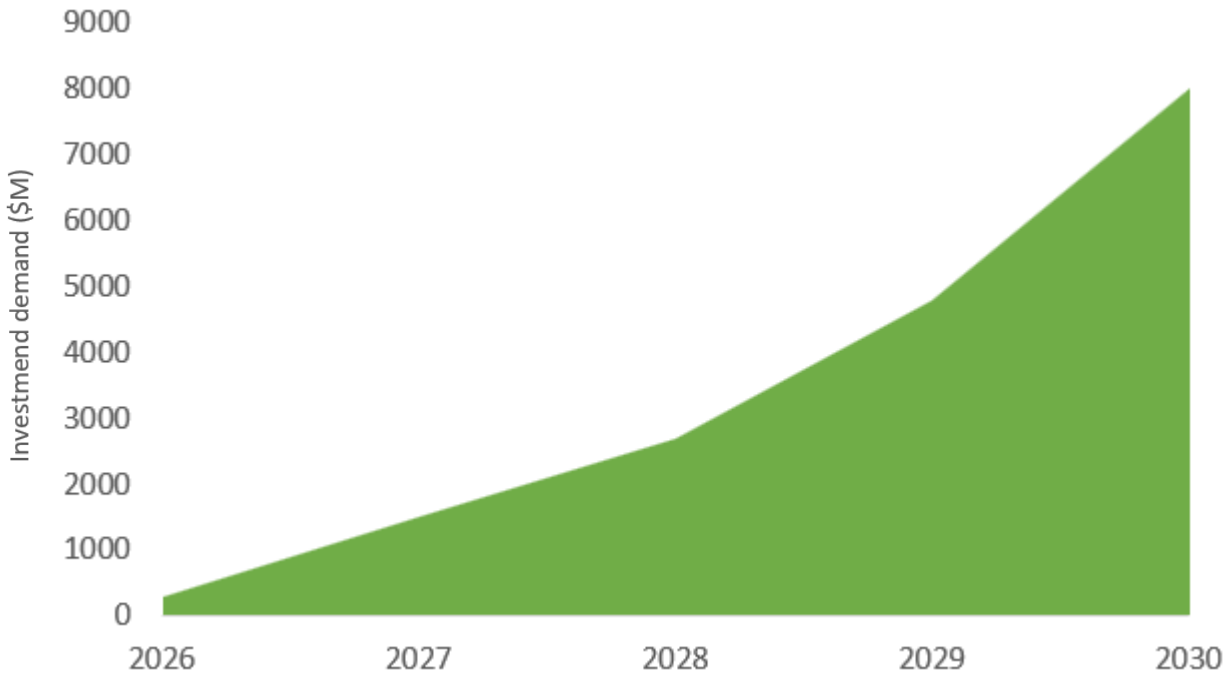


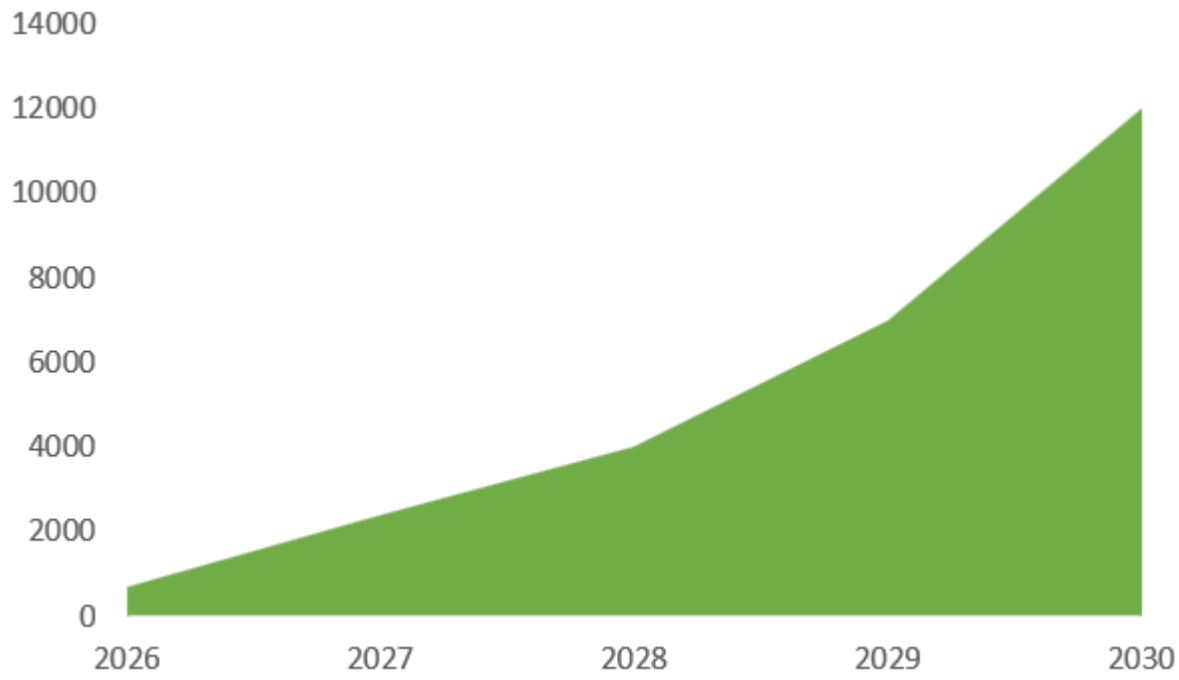
EXHIBIT 4

Year	2026	2027	2028	2029	2030
Investment Demand ($M)	500	1950	3350	5900	10000
Year on year growth	na	290%	71%	76%	69%

CAGR (Green H2 Export Demand)	**20%** *(based on SAREP analysis)*
Import Demand met by 2030	**8%** *(taking an average of business's 6% & NHM's 10%)*



Business as usual (6% import demand met)



NHM Target (10% import demand met)

EXHIBIT 5

Carbon Emissions avoided from Green H2 by 2030		Rationale
Taking 3 phases for capturing Green H2 demand		
Phase 1(2025-2026)	20%	
Phase 2(2027-2028)	50%	
Phase 3(2029-2030)	70%	
Estimated demand based on NHM targets(MMT)		
2025	0.40	
2026	0.85	
2027	1.45	
2028	2.15	
2029	3.20	
2030	5.00	
Total Estimated Demand	13.05	*based on NHM Targets*
Green H2 demand captured based on above assumptions		
2025	0.08	
2026	0.17	
2027	0.73	
2028	1.08	
2029	2.24	
2030	3.50	
Total Demand Captured	7.79	*based on SAREP reports*
CO2 Saved by 2030		
C02 emissions for 1MMT of grey hydrogen produced	12.4	https://gh2.org/blog/mirage-blue-hydrogen-fading
Captured Green H2 demand	7.79	MMT
Carbon Emissions Reduced for 7.79 MMT	**96.5**	MMT


IIMB
तेजस्वि नावधीतमस्तु
THANK YOU